

- Q.29 Explain addressing modes of 8085. (CO9)
- Q.30 Write differences between static and dynamic memories. (CO8)
- Q.31 Write advantages of 16 bit microprocessor and over 8 bit. (CO9)
- Q.32 Draw the block diagram of full adder circuit and also draw its truth table. (CO4)
- Q.33 Solve the following Boolean expression. (CO3)
 $Y = AB + A(B+C) + B(B+C)$
- Q.34 Write short note on display devices (CO4)
 a) LED Display b) 7-Segment Display
- Q.35 Define distributive law and commutative law. (CO3)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Reduce the Boolean expression by using K-map. Also draw the logic circuit. (CO3)
 $Y = \sum m(0,1,4,5,12,13,8,9,2,6,14)$
- Q.37 Draw and explain architecture of 8085 microprocessor. (CO9)
- Q.38 Explain J-K Flip Flop with truth table. (CO5)

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Time : 3Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Which logic gate produces an output of 1 only when all of its inputs are 0. (CO2)
 a) AND GATE b) OR GATE
 c) XOR GATE d) NAND GATE
- Q.2 How many bits make a byte. (CO1)
 a) 4 bits b) 8 bits
 c) 16 bits d) 2 bits
- Q.3 Expand LCD (CO4)
 a) Light cell device b) Liquid crystal display
 c) Light crystal display d) None
- Q.4 If J=0 and K=1 then the output Q_____. (CO5)
 a) 1 b) 0
 c) No change d) Toggle
- Q.5 A 3 variable K-map has _____ cells. (CO3)
 a) 9 b) 16
 c) 8 d) 3

Q.6 _____ Pin of the 8085 is used to provide power supply voltage. (CO9)

- a) VCC b) VDD
- c) RESET d) CLK

Q.7 Shift register is a combination of _____. (CO6)

- a) Flip-Flop b) Encoders
- c) Decoders d) Counters

Q.8 8085 microprocessor is _____. (CO9)

- a) 40 Pin b) 60 Pin
- c) 35 Pin d) None of these

Q.9 The inputs of a NAND gates are connected together. The resulting circuit is _____. (CO2)

- a) OR GATE b) AND GATE
- c) NOT GATE d) None of these

Q.10 How many select lines does a 4:1 multiplexer have. (CO4)

- a) 1 b) 2
- c) 3 d) 4

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 PIPO stands for _____. (CO6)

Q.12 IC used for NOT Gate is _____. (CO2)

Q.13 Write the name of any one type of A/D converter. (CO7)

Q.14 Outputs of half adder are _____ and _____. (Co4)

Q.15 $A+1 =$ _____ (CO3)

Q.16 Maximum number of input lines in a 8:1 MUX are _____. (CO4)

Q.17 ALU stands for _____. (CO9)

Q.18 Give one example of data transfer instruction. (CO9)

Q.19 Draw symbol of XOR gate. (CO2)

Q.20 Add $1010 + 1110 =$ _____ (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Covert $(467.82)_8$ into binary. (CO1)

Q.22 Explain R/2R D/A converter (CO7)

Q.23 Explain NAND gate as universal gate. (CO2)

Q.24 Write four applications of Flip-Flop. (CO5)

Q.25 Compare multiplexer and demultiplexer. (CO4)

Q.26 Draw the truth table of SR Flip-Flop and explain its operation (CO5)

Q.27 Explain in brief SIPO shift register. (CO6)

Q.28 Subtract 1101 from 1001 using 2's complement method. (CO1)